
KV-Head Placement in a Width-Scaled Language Model

A Fixed-Time Follow-up under Local-Global Attention

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Abstract

Multi-query and grouped-query attention reduce key/value redundancy, but they usually assign one KV-head count uniformly across depth. We ask whether a width-scaled language model should spend full multi-head KV projections where its attention is global. After a width-512 pilot fell below our practical effect threshold, we repeat the same pinned five-minute single-A100 pipeline at width 640. We compare all-MHA, uniform MQA, and two hybrids that match KV-head budget, KV-dependent parameters, value-embedding parity, and optimizer shape multiset, but not layer depth or per-step attention compute. Global allocation beat matched Local by 0.001870 val_bpb, meeting the 0.001 threshold. The unchanged Global rerun reached 1.107559. The 0.002041 unchanged-run gap exceeds the 0.001870 placement contrast. Thus the threshold is met, but the post-pilot design and single run per placement preclude a robust scale-law or statistical claim.

1. Introduction

Multi-query attention (MQA) shares one key/value projection across query heads to reduce decoder memory traffic (Shazeer, 2019). Grouped-query attention (GQA) interpolates between MQA and multi-head attention (MHA), often recovering quality with more KV heads (Ainslie et al., 2023). These designs normally use the same sharing ratio in every layer, even when attention scope varies with depth. Local-global hybrids are already practical—Gemma 2, for example, combines interleaved local/global attention with GQA (Gemma Team, 2024).

Efficiency work has exposed several distinct allocation

axes. Cost-optimal GQA varies a model-wide head configuration with context length and model size (Chen et al., 2025); layerwise KV-cache methods vary how many past tokens are retained during inference (Wang et al., 2024); and ConSA learns where to place full versus sliding-window attention (Chen et al., 2026). We instead hold the attention-window schedule fixed and redistribute the number of learned KV heads across layers during pretraining.

We study the eight-layer SSSGSSSG schedule in the pinned autoresearch benchmark (Karpathy, 2026; VESSL AI, 2026). The central question is narrow: under a fixed five-minute budget, does putting full MHA KV projections in global layers lower val_bpb relative to an equal KV-parameter-budget placement at value-embedding-matched local layers? Layer depth and per-step attention compute remain unmatched. The motivating intuition is that query heads attending over the full context may benefit more from distinct key/value subspaces than heads restricted to a local window. This is a hypothesis, not an assumption of the analysis.

Our contribution is threefold: an implementation-audited scale-up of a matched contrast; a complete fixed-time control table; and an unchanged winner rerun. The 0.001 practical threshold and selection rule carry forward from the pilot, but width 640 was chosen after seeing it, making this a transparent follow-up rather than an independent preregistration.

2. Method

Data and metric. We use cookbook commit 97a0af14b0a (the full pin is recorded in the evidence manifest), whose prepare.py and baseline train.py match the named Karpathy source pin. Preparation downloads ten training shards from ClimbMix-400B-Shuffle plus fixed validation shard 6542 and trains the pinned 8,192-token BPE on the training split. All conditions use the same 2,048-token context and evaluate exactly 20,971,520 validation tokens. Validation bits per byte sums token cross-entropy in nats, divides by the number of target bytes, and converts to base two; byte-less special tokens are excluded. This frozen val_bpb (lower

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Table 1. Matched scale-up design. S/G denote a 1,024-token local window and 2,048-token global context; H/M denote five/one KV heads. Y/N marks layers with/without a value embedding. Both hybrids use 16 KV-head units and 15,204,736 KV-dependent parameters.

Row	Zero-indexed layer								Matched budget	
	0	1	2	3	4	5	6	7	KV units	KV-dep. P
Scope	S	S	S	G	S	S	S	G	–	–
Value embedding	N	Y	N	Y	N	Y	N	Y	–	–
Global placement	M	M	M	H	M	M	M	H	16	15,204,736
Local placement	M	H	M	M	M	H	M	M	16	15,204,736

is better) is the sole primary metric.

Model and optimization. The decoder has eight layers, width 640, five query heads of width 128, and the window schedule SSSGSSSG; S uses a 1,024-token window and G sees the full 2,048-token context. Odd layers also contain ResFormer value embeddings. We retain the pinned MuonAdamW recipe: Muon for matrix parameters, AdamW for embeddings and scalars, a 524,288-token global batch, bfloat16 autocast, seed 42, and matrix/embedding/unembedding/scalar learning rates of 0.04/0.6/0.004/0.5. Device batch 64 with four-way accumulation preserves the 524,288-token global batch. Muon weight decay starts at 0.2, Adam betas are 0.8/0.95, there is no warmup, and learning rates linearly decay over the final half of training. Every run receives 300 timed training seconds after compilation warmup; script-level time additionally includes model setup, compilation, and final evaluation. Completed tokens, steps, script time, parameters, and peak memory are diagnostics, not additional objectives.

Interventions. The model has five query heads per layer. Let H denote five KV heads and M one KV head. We run an untouched all-MHA baseline (HHHHHHHH), uniform MQA (MMMMMMMM), global-only MHA at layers 3 and 7 (MMHMMMH), and local-only MHA at layers 1 and 5 (MHMMMHMM). Layers are zero-indexed. The two hybrids each have two five-KV and six one-KV layers. Crucially, the baseline gives ResFormer value embeddings to odd layers; both pairs therefore place their two full-KV layers on the same value-embedding parity.

For layer ℓ with h_ℓ KV heads, the audited KV-dependent count is

$$P_{KV} = \sum_{\ell=0}^7 [2dh_\ell d_h + \mathbb{K}_{\ell \text{ odd}} (Vh_\ell d_h + 32h_\ell)],$$

with $d = 640$, $d_h = 128$, and vocabulary $V = 8192$. Both hybrids therefore have exactly 58,458,512 total parameters. A meta-device audit of the frozen model definition gives identical state-shape multiset fingerprint 81354028ba50 and optimizer-group/shape fingerprint f2477d093660 for the two hybrids. Their source files differ only in the pattern constant; the audit record retains full SHA-256 values.

Execution and provenance. Every scientific Job clones and verifies the pinned cookbook commit and the pre-overwrite baseline Git blob, then writes an archived, SHA-256-verified `train.py` payload. Relative to the official baseline, every condition shares only width ratio 64→80 and device batch 128→64; candidates additionally implement the declared KV mask. No other benchmark file is changed. Every run uses the same single A100-SXM4-80GB, container image, prepared cache, evaluation fingerprint, seed, and sequential execution protocol. The global rerun uses the exact same `train.py` SHA-256 as its planned run. This direct payload route is disclosed because a writable participant fork was unavailable.

Decision rule. We retain the pilot’s placement contrast

$$\Delta_{\text{place}} = \text{val_bpb}_{\text{local}} - \text{val_bpb}_{\text{global}}.$$

We report the contrast as meeting the carried-forward practical threshold only when $\Delta_{\text{place}} \geq 0.001$; this is not a statistical significance test. A smaller magnitude is descriptive, and $\Delta_{\text{place}} \leq -0.001$ meets the reverse-direction threshold. Candidates run sequentially. A candidate is kept only if it lowers `val_bpb`; an overall improvement claim additionally requires an unchanged rerun below the same baseline. With one run per planned condition, we report observations and do not attach sampling-based significance claims.

3. Results

Primary placement test. Global reached 1.109600 and matched Local reached 1.111470, so $\Delta_{\text{place}} = 0.001870$. This exceeds the carried-forward 0.001 threshold by 0.000870 and supports the directional placement hypothesis in this follow-up. The jobs are closely matched—302 versus 301 steps, 158.3M versus 157.8M tokens, equal 58.5M parameters, and 25,413.1 versus 25,413.9MB peak allocation—but a single pair is not an uncertainty estimate.

The unchanged Global rerun reached 1.107559. Its absolute difference from the planned Global run is 0.002041 `val_bpb`. This checks the selected source against the same baseline but neither repeats Local nor estimates a sampling distribution for the primary contrast. Because 0.002041 exceeds the 0.001870 placement gap, the direc-

Table 2. Width-640 single-A100 outcomes. Δ is baseline minus condition; positive is better.

Condition	Pattern	bpb $\downarrow$	Δ	P (M)	Steps	Tokens (M)	Script (s)	VRAM (MB)
All-MHA	HHHHHHHH	1.115715	0.000000	70.8	280	146.8	387.4	28097.9
Uniform MQA	MMMMMMMM	1.113916	0.001799	48.8	314	164.6	386.8	24302.6
Global	MMMHMMMH	1.109600	0.006115	58.5	302	158.3	390.6	25413.1
Local	MHMMHMM	1.111470	0.004245	58.5	301	157.8	390.4	25413.9
Confirmation	MMMHMMMH	1.107559	0.008156	58.5	306	160.4	389.5	25413.1

tional threshold result should not be read as a robustness claim.

Baseline and throughput controls. All reduced-KV candidates beat the all-MHA baseline of 1.115715: Global by 0.006115, Local by 0.004245, and uniform MQA by 0.001799. Under the retained selection rule, only the lowest Global condition was rerun. A below-baseline rerun is a *protocol-level confirmation*, not a multi-seed claim.

Uniform MQA completed 314 steps and 164.6M tokens, versus 280 steps and 146.8M tokens for all-MHA, and improved by 0.001799. Global improved more despite fewer steps than MQA, but these comparisons are neither parameter- nor token-matched and cannot identify whether capacity, throughput, or optimization caused the gains.

Relation to the pilot. At width 512, the same masks yielded Global 1.107673 and Local 1.108214, a 0.000541 contrast below threshold; the unchanged Global repeat gap was 0.001246. The present contrast is $3.46\times$ larger. Because width 640 was selected after the pilot, this is evidence of scale sensitivity, not a fitted scaling law.

4. Reproducibility and Audit

All scale-up Jobs match the exact evaluation fingerprint, prepared cache, and one-A100 resource specification. Raw logs, executed sources, Job metadata, configs, and the shape audit form evidence bundle `7c8991de4535`. The scientific set contains the baseline, three candidates, and conditional rerun; no scale-up outcome is omitted.

5. Analysis and Discussion

Why Global Scope May Need Head Diversity. Query heads attending over the entire context must resolve long-range dependencies and multi-hop reasoning, which conceptually require distinct, specialized key-value subspaces. In contrast, local attention largely focuses on n-gram statistics and syntactic locality, which may tolerate a lower-rank KV representation (as forced by uniform MQA). Future work will analyze attention-head similarity and representation rank to verify if global layers exhibit higher intrinsic dimensionality.

What the scale comparison does and does not show.

The placement contrast grows from 0.000541 at width 512 to 0.001870 at width 640 while the two scale-up hybrids retain almost identical realized throughput. One plausible interpretation is that distinct KV subspaces become more useful for full-context retrieval once the residual stream and query-head count grow. An equally compatible explanation is scale-specific optimization noise: the fixed seed does not eliminate nondeterministic kernels, and neither placement was repeated as a pair. We therefore treat the ratio between contrasts as a design signal for the next study, not as an effect-size estimator.

Run Variance and Cache Latency. The fixed-time protocol exposes capacity tradeoffs, but mechanism requires broader placement ablations and actual cache/latency analysis. Multi-seed learning curves are needed to separate architecture from optimization noise; here the 0.002041 same-source gap exceeds the cross-placement contrast.

Related Work. Recent studies explore layerwise KV-cache management (Wang et al., 2024) and learnable attention scope allocation (Chen et al., 2026). We extend these directions by investigating heterogeneous KV-head allocation across fixed local-global attention schedules.

6. Limitations and Next Test

Validation bpb after five timed minutes does not establish downstream quality, convergence, decoding latency, or realized KV-cache savings. The all-MHA model is 70.8M parameters, but the primary hybrids are 58.5M; “larger” here means a 40% scale-up from the corresponding 41.7M pilot hybrids, not a modern large language model. Device batch was reduced to fit conservatively, preserving global batch tokens through accumulation but changing the microbatch from the pilot. The final layer is global by construction, so scope and depth cannot be fully crossed by merely moving KV heads.

The decisive next test is a paired multi-seed 2×2 study over widths 512 and 640, repeating both Global and Local masks plus all-MHA baselines. A second window schedule should swap local/global scope at matched depths, and a token-matched report should accompany fixed-time outcomes. Intermediate two- or three-KV-head GQA would

Table 3. Claim audit for the width-640 follow-up. Verified controls narrow specific alternatives; the residual column states the remaining claim limit.

Threat	Verified control/evidence	Residual limitation
Implementation mismatch	The hybrids match KV units, KV-dependent and total parameters, value-embedding parity, state shapes, and optimizer-group shapes; sources differ only in the mask constant.	Identical tensor shapes occur at different depths and scopes, so scope remains coupled to within-block depth.
Fixed-time confounding	Both hybrids use 300 timed seconds and differ by one step, 0.5M tokens, and 0.8MB peak allocation.	Baseline and MQA comparisons are not parameter- or token-matched; theoretical attention work is not forced equal across scope placements.
Scale comparison	Depth, context, masks, data, seed, optimizer, metric, and A100 class match the width-512 pilot; width and its safety microbatch change are explicit.	Only two widths are observed, and width 640 was chosen after seeing the pilot; this cannot identify a scaling curve.
Run variation	The selected source is rerun byte-for-byte under the same resource and cache.	Only the winner is repeated; the primary Global-Local contrast has no variance estimate or confidence interval.
Provenance drift	Commit, official blob, payload SHA-256, cache, evaluation, source, config, log, and Job identities are archived.	Direct payload delivery replaces a participant fork, and one data mixture and hardware class limit external validity.

then test whether the observed benefit is a placement effect or simply the consequence of the chosen capacity endpoints.

7. Conclusion

At width 640, allocating the two full-KV layers to global attention beats the matched local placement by 0.001870 and clears the carried-forward threshold. This differs from the below-threshold width-512 pilot without changing the direction, suggesting scale sensitivity. One run per placement, a post-pilot scale choice, fixed-time confounding, and coupling of scope with depth prevent a causal or robust scaling claim; the same-source gap is larger than the placement contrast. Paired multi-seed replication at both widths is the next test. Within this protocol, scope-aligned placement is supported at width 640; generalization and mechanism remain open.

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